layer.

Sean Lynch · Vercel Design Engineer take-home · 9 October 2026

A considered place to practise miniature painting. Painting is a personal interest, and I wanted to explore a companion for the moments around the miniature: choosing a palette, understanding the next mark and returning to an unfinished study. Layer brings those decisions into one continuous workflow, with materials, demonstrations, contextual help and saved notes close to the work.

Designing through Foundry. I used my Foundry design skill to guide the site's design and refinement, alongside AI-assisted implementation. I directed the hierarchy, typography, spacing, tone and interactions through repeated browser review. The important change was restraint: oversized illustrated brushes gave way to quiet white surfaces where the paint itself demonstrates the technique. Controls sit inside the stage; optional detail stays out of the primary reading path. The same visual language carries through the guides, equipment and technique library.

Vercel AI Gateway, with a deliberate boundary. The Workbench coach uses the Vercel AI SDK and AI Gateway to match a question to reviewed guidance for the current step. Structured output selects an allowed note; Layer supplies its exact teaching text, sources and an authored “Show me” demonstration. A live model response had reversed advice about excess paint, so I narrowed its role from writing instruction to selecting guidance. AI cannot invent a technique, grade the miniature or complete a step. Reviewed fallback guidance keeps the lesson useful when AI is unavailable.

Make the workflow dependable. Next.js, Tailwind and TypeScript support a responsive app deployed on Vercel. Deterministic paint studies share one playback clock, with Before, Application and Result checkpoints, adjustable paint amount and a fixed reference. WebGL has a readable SVG fallback. Only the learner advances practice; Next has Undo, notes reopen at their original step, and saved progress survives reload. The interactive pauldron adds a movable palette reference while keeping lesson progression separate.

Tradeoffs and assumptions. Device-local saving removes signup friction but gives up cross-device continuity. Digital colours and paint coverage are illustrations, not calibrated matches or fluid simulation. The catalogue grew beyond the initial foundational scope; one complete Ultramarines journey is the clearest demonstration. I am testing the hypothesis that keeping these decisions together helps practice, not claiming a measured learning outcome.

Next. Desk-test one lesson with a beginner, evaluate AI note-selection relevance, and add durable usage limits before broader release. Prioritise that evidence over more catalogue breadth.

Delivery: Application · GitHub · Active time: **5 hours 36 minutes**.